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Status	Beendet
Beendet am	Mittwoch, 31. Januar 2024, 16:58
Verbrauchte Zeit	1 Stunde
Bewertung	50,00 von 50,00 (100%)

Frage 1

Vollständig

Nicht bewertet

General information:

- Not counting this first "task" (i.e. the statutory declaration below), there are 12 problems with 50 points in total.
- Read all problems and possible answers carefully and take your time to think about them.
- Multiple-choice questions: Choosing wrong answers leads to deduction of points within that question. However, you cannot get less than 0 points on a question.
- Numerical questions: Wrong answers do not lead to point deductions.
- For numerical answers, use "." for the decimal point. In case you get an error message or the question is marked as not answered, try using "," instead.
- When it says that you can use a pocket calculator, you may also use the calculator of your operating system.
- After some questions, there is an optional text field. This field will not influence the automatic grading. Here, you can write down subresults of your calculations. It may be used to your advantage at the end of the semester. There will be no other form of submitting information, i.e., no emails or Moodle uploads with photos or pdfs after the exam.
- You can ask questions during the exam only via the chat window and only directly to the exam supervisor. Your multiple-choice answers may be displayed in random order. So, don't refer to them by their label.
- You have 60 minutes for the exam. After this time is expired, your exam will automatically be submitted. There is no overtime.
- Once you have submitted your exam, please state so in the chat window, e.g. write "submitted", don't wait for a reply, and then just leave the Zoom meeting.
- Strictly forbidden: All forms of audio or video recording by the candidates during the examination. This includes, but is not limited to, screenshots, downloads and any copy of the exam content (digital or analogue). Also after the exam, you are not allowed to copy or share or pass on the exam material in any way whatsoever.
- The Zoom session of the exam may be recorded. (By law, recordings must not be kept longer than 72 hours. We will delete it even earlier in case everything runs smoothly, which is what we expect.)

Statutory declaration:

With clicking on "I agree" below, you agree to this statutory declaration ("Eidesstattliche Erklärung"): "I hereby confirm that I complete this closed-book exam on my own and only using the allowed materials, in particular without any written or electronic notes and without any help from or communication with other people. I understand that violation of this declaration may cause legal consequences."

Wählen Sie eine Antwort:

- ☒ I agree
- ☐ I do not agree

Frage 2

Richtig

Erreichte Punkte 4,00 von 4,00

Which statements about ensemble methods, in particular random forests and (gradient) boosting, are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. A random forest in general has smaller bias than the individual trees.
- ☒ b. A common measure of variable importance in random forests is the mean decrease in Gini impurity. ✓
- ☐ c. In AdaBoost, the less accurate the weak classifiers are, the higher the influence they are given for the classification output.
- ☒ d. In a random forest, each individual decision tree is built upon a random bootstrap subsample of the original data set. ✓
- ☐ e. Gradient Boosted Trees are generally easier to tune compared to Random Forests.
- ☒ f. Forward stagewise additive modelling is a robust boosting method which can be applied to different loss functions. ✓
- ☐ g. In Boosting, the main idea is that those samples which led to a small error (i.e. were classified well) in an iteration get a larger weight in the next iteration.
- ☐ h. In general, ensemble methods aim at reducing the bias and increasing the variance.

Frage 3

Richtig

Erreichte Punkte 4,00 von 4,00

Boosting:

For a given data set, an algorithm has found a sequence of weak binary classifiers g_n with $n = 1, \dots, 6$.

For a data point $\mathbf{x}$, the outputs of the individual weak classifiers and their respective weight factors are:

$$g_1(\mathbf{x}) = -1, \quad \alpha_1 = 4.5$$

$$g_2(\mathbf{x}) = +1, \quad \alpha_2 = 0.5$$

$$g_3(\mathbf{x}) = +1, \quad \alpha_3 = 6.5$$

$$g_4(\mathbf{x}) = -1, \quad \alpha_4 = 4.5$$

$$g_5(\mathbf{x}) = +1, \quad \alpha_5 = 1.0$$

$$g_6(\mathbf{x}) = +1, \quad \alpha_6 = 0.5$$

What is the output of the binary strong (boosted) classifier?

Antwort: -1



Frage 4

Vollständig

Nicht bewertet

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$$g_1 \cdot a_1 + g_2 \cdot a_2 + \dots + g_6 \cdot a_6 = -4.5 + 0.5 + 6.5 - 4.5 + 1 + 0.5 = -0.5$$

$$\text{sign}(-0.5) = -1$$

Frage 5

Richtig

Erreichte Punkte 4,00 von 4,00

Which statements about Logistic Regression (and related topics) are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. In Logistic Regression, minimization of the loss function can be achieved by gradient descent. ✓
- ☐ b. Logistic Regression is a non-convex problem as can be seen by the positive-semidefiniteness of the Hesse matrix.
- ☐ c. The logistic function reads: $\sigma(z) = (1 - \exp(z))^{-1}$. It is also known as the sigmoid function.
- ☐ d. Assume that y data are generated around a function $g(x)$ with Gaussian noise. Maximizing the negative log-likelihood of observing the data is equivalent to minimizing the squared error between y and $g(x)$.
- ☐ e. The likelihood function for a Bernoulli distribution reads: $\mathcal{L}(\{\mathbf{x}, y\}; \mathbf{W}) = \prod_{i=1}^l [g(\mathbf{x}_i; \mathbf{W})^{y_i} + (1 - g(\mathbf{x}_i; \mathbf{W}))^{1-y_i}]$.
- ☒ f. The momentum term in gradient descent is particularly helpful in flat regions of the loss function. ✓
- ☐ g. Logistic regression has a closed-form solution.
- ☒ h. The softmax function is a generalization of the sigmoid function and is suitable for multi-class classification. It is also known as the Boltzmann distribution. ✓

Frage 6

Richtig

Erreichte Punkte 4,00 von 4,00

Cross Entropy Loss in Logistic Regression:

Assume you have two data samples:

$$\mathbf{x}_1, y_1 = 1$$

$$\mathbf{x}_2, y_2 = 0$$

The model predictions are:

$$g(\mathbf{x}_1) = 0.9$$

$$g(\mathbf{x}_2) = 0.4$$

Compute the Cross Entropy Loss.

Reminder: Use the natural logarithm $\ln = \log_e$ in your calculation, not a logarithm to some other base.

Note: Do not take the average over the two samples (as may also be a common convention), but follow the convention from the lecture where we take the sum over data points (without normalization).

You can use a pocket calculator. Give the answer as a real number, rounded to two decimal places. Use "." for the decimal point. Number format example: 1.23

Antwort:

**Frage 7**

Vollständig

Nicht bewertet

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$$-(y_1 \ln(g_1) + (1-y_1) \ln(1-g_1)) - (y_2 \ln(g_2) + (1-y_2) \ln(1-g_2)) = -1 \ln(0.9) - 1 \ln(1-0.4) = 0.62$$

Frage 8

Richtig

Erreichte Punkte 4,00 von 4,00

Which statements about Perceptrons and Perceptron Learning are true?

Notation as in the lecture:

Weight vector: $\mathbf{w}$

Threshold: θ

Data samples: $\mathbf{x}_i$

Learning rate: $\eta > 0$

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. In the learning algorithm: If $g(x_i; \mathbf{w}, \theta) = 0$ and $y_i = 1$, then the update rule for the weight vector reads: $\mathbf{w} := \mathbf{w} + \eta \mathbf{x}_i$. ✓
- ☒ b. If the data set is linearly separable, then there exists a stopping criterion such that with probability 1 the perceptron learning algorithm will terminate and solve the learning task. ✓
- ☐ c. In the learning algorithm: If $g(x_i; \mathbf{w}, \theta) = 1$ and $y_i = 0$, then the update rule for the threshold reads: $\theta := \theta - \eta$.
- ☐ d. A perceptron can solve the XOR problem, but only in certain instances (i.e. not with probability 1) when the algorithm terminates.
- ☐ e. The final solution (values of the $\mathbf{w}$ and θ) of the perceptron learning algorithm is always unique.
- ☒ f. The perceptron is a simple linear threshold unit. ✓

Frage 9

Richtig

Erreichte Punkte 4,00 von 4,00

Which statements about Artificial Neural Networks are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

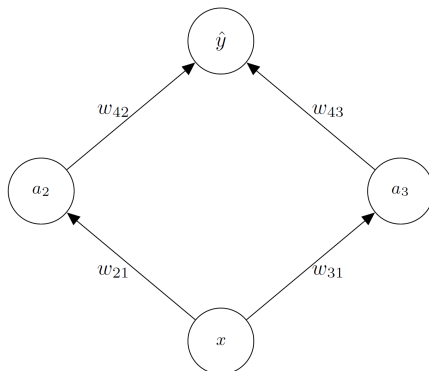
- ☐ a. Double descent is the phenomenon that the test risk first decreases with increasing number of parameters, then increases up to the interpolation threshold, and then decreases again. The final decrease takes place in the so-called over-parametrized regime, where there are more data points than parameters.
- ☒ b. Multi-layer perceptrons are equivalent to fully connected feed-forward neural networks. ✓
- ☐ c. Let us denote the learning rate with $\eta > 0$, the weights with $\mathbf{w}$, and the empirical risk with R . Then the gradient descent update rule reads: $\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} + \eta \nabla_{\mathbf{w}} R$.
- ☒ d. The Universal Approximation Theorem guarantees that neural networks can in principle approximate arbitrarily well any continuous function $f \in [0, 1]^m$. ✓
- ☒ e. Without activation functions, neural networks are linear functions with respect to their weights. ✓
- ☐ f. Using the matrix-vector notation from the lecture (Unit 6), where $l = 0$ denotes the input layer, the recursive formula for delta errors is given by: $\delta^{[l+1]T} = \delta^{[l]T} \mathbf{J}^{[l]}$. Here, $\mathbf{J}^{[l]}$ is the single-layer Jacobian for layer l .
- ☐ g. Dropout and weight decay are regularization techniques which aim at avoiding underfitting.
- ☐ h. The vanishing gradient problem can be avoided by the use of suitable loss functions.

Frage 10

Richtig

Erreichte Punkte 5,00 von 5,00

Consider the following neural network:



The input unit 1 has activation $a_1 = x$. The preactivations of the hidden units 2 and 3 are denoted as s_2 and s_3 , and their activations are a_2 and a_3 , respectively. The preactivation of the output unit 4 is denoted as s_4 and the output as $a_4 = \hat{y}$. The true label is denoted as y . There are no biases.

In the hidden layer, we use Relu as activation function, i.e. $f_2(s) = f_3(s) = \text{Relu}(s) = \max(0, s)$. In the output layer, the activation is the identity function.

The delta-error at the output is denoted as δ_4 , and the hidden delta-errors as δ_2 and δ_3 , respectively. As loss function, we use the squared error, i.e. $L(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$.

The learning rate is denoted as η . The network weights are updated via backpropagation and gradient descent, and these updated weights are denoted by superscript "new".

We are given the input value, the current network weights, the true label, and the learning rate for gradient descent update:

$$x = 3$$

$$w_{21} = 2$$

$$w_{31} = -2$$

$$w_{42} = -\frac{1}{3}$$

$$w_{43} = \frac{3}{2}$$

$$y = 1$$

$$\eta = \frac{1}{20}$$

Compute the following quantities (forward pass and backpropagation):

- $\hat{y} =$ ✓
- $\delta_2 =$ ✓
- $\frac{\partial L}{\partial w_{21}} =$ ✓
- $w_{21}^{\text{new}} =$ ✓

You can use a pocket calculator. Give the answers as real numbers, rounded to two decimal places. Use "." for the decimal point. Number format example: 1.23

Frage 11

Vollständig

Nicht bewertet

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$$s_2 = w_{21} * x = 6 \rightarrow a_2 = 6$$

$$s_3 = w_{31} * x = -6 \rightarrow a_3 = 0$$

$$y_{\text{dach}} = s_4 = a_4 = w_{42} * a_2 + w_{43} * a_3 = w_{42} * a_2 = -2$$

$$\delta a_4 = dL/ds_4 = s_4 - y = -2 - 1 = -3$$

$$\delta a_2 = f'(s_2) * \delta a_4 * w_{42} = 1 * (-3) * (-1/3) = 1$$

$$dL/dw_{21} = \delta a_2 * a_1 = 1 * 3 = 3$$

$$w_{21_new} = w_{21_old} - \eta * \delta a_2 * a_1 = 2 - 1/20 * 1 * 3 = 1.85$$

Frage 12

Richtig

Erreichte Punkte 4,00 von 4,00

Consider a neural network in matrix-vector notation.

Assume that the preactivation vector for layer $l - 1$ is given by $\mathbf{s}^{[l-1]} = (-2, -1, 3)^T$. As activation function f take Relu.

The weight matrix reads $\mathbf{W}^{[l]} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}$.

Compute the bottom right entry of the single-layer Jacobian $\mathbf{J}^{[l]}$, i.e. $J_{3,3}^{[l]}$.

The answer is an integer number.

Antwort:



Frage 13

Nicht beantwortet

Nicht bewertet

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Frage 14

Richtig

Erreichte Punkte 4,00 von 4,00

Convolutional Neural Networks (CNNs):

Which statements are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Re-using a kernel weight matrix multiple positions of an image is called weight sharing. ✓
- ☒ b. With kernel weight matrix $w_{i,j}$, kernel size R , and input image $x_{a,b}$, the feature map preactivation is calculated by $s_{a,b} = \sum_{i=0}^{R-1} \sum_{j=0}^{R-1} w_{i,j} x_{a+i,b+j}$. ✓
- ☒ c. Dataset augmentation can help to overcome the problem of limited quantity and/or diversity of given data. ✓
- ☐ d. AlexNet is a famous CNN, whose good learning performance is due to skip connections.
- ☐ e. With input dimension A , kernel size R , padding P , and stride S , the feature map dimension is $(A + R + 2P)/S + 1$.
- ☐ f. Each convolutional layer always uses exactly one kernel.
- ☒ g. Transfer Learning allows to use pretrained kernels from other networks which were trained on different data sets. ✓
- ☐ h. Pooling decreases the receptive field through the depth of the network.

Frage 15

Richtig

Erreichte Punkte 4,00 von 4,00

Convolution:

Consider the input image

$$\mathbf{x} = \begin{pmatrix} 2 & 1 & 2 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 2 & 0 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 1 \\ 1 & 2 & 3 & 2 & 3 & 2 & 1 \end{pmatrix}$$

and the filter

$$\mathbf{W} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Assuming stride $S = 1$ and no padding, compute the bottom left entry $s_{5,1}$ of $\mathbf{s} = \mathbf{W} * \mathbf{x}$.Assuming no padding and stride $S = 2$, the dimension of the output matrix becomes $d \times d$. Compute d .Assuming (symmetric, as convention in the lecture) padding $P = 2$ and stride $S = 2$, the dimension of the output matrix becomes $D \times D$. Compute D .

- $s_{1,5} =$ ✓
- $d =$ ✓
- $D =$ ✓

Frage 16

Nicht beantwortet

Nicht bewertet

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Frage 17

Richtig

Erreichte Punkte 4,00 von 4,00

Recurrent Neural Networks (RNNs):

Which statements are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

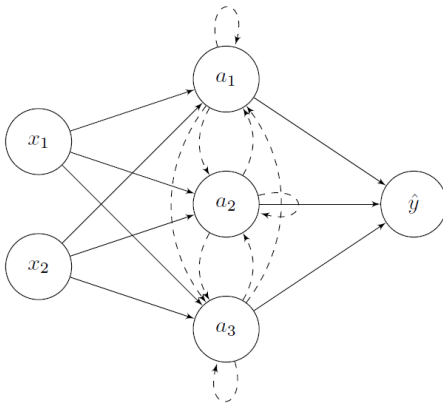
- ☐ a. The constant error carousel in LSTMs uses a non-linear activation function in order to avoid/mitigate the vanishing gradient problem.
- ☒ b. We denote with $\mathbf{R}$ the recurrent weight matrix, with f the activation function, and with $\mathbf{s}(t-1)$ the preactivation at time $t-1$. For stable gradients, we need $\|\mathbf{R} \text{diag}(f'(\mathbf{s}(t-1)))\| \approx 1$. ✓
- ☒ c. RNNs can be trained via backpropagation through time. ✓
- ☐ d. In the backward pass, the gradient of the loss function L with respect to the input weight matrix $\mathbf{W}$ reads:
$$\partial L / \partial \mathbf{W} = \sum_{t=1}^T \delta(t) \mathbf{x}(t-1)^T$$
- ☐ e. LSTMs use gates to store information in a constant error carousel.
- ☐ f. In an RNN, each unit in a hidden layer can only be connected to itself and the directly neighboring unit in the layer.
- ☒ g. RNNs are Turing complete. ✓
- ☒ h. RNNs use weight sharing in the sense that the network is slid over the input sequence and the hidden-layer weight matrix is reused for all timesteps. ✓

Frage 18

Richtig

Erreichte Punkte 5,00 von 5,00

Consider the following recurrent neural network:



The activation function in both the hidden layer and in the output layer is Relu. The initial activation vector in the hidden layer is set to $\mathbf{a}(t=0) = (0, 0, 0)^T$.

The input vector at the first time step is $\mathbf{x}(t=1)$. The input vector at the next time step is $\mathbf{x}(t=2)$. The input weight matrix is denoted as $\mathbf{W}$. The recurrent weight matrix is denoted as $\mathbf{R}$. The output weight matrix is denoted as $\mathbf{V}$.

We are given:

$$\mathbf{x}(t=1) = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\mathbf{x}(t=2) = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\mathbf{W} = \begin{pmatrix} -1 & 3 & 1 \\ 2 & 1 & -1 \end{pmatrix}$$

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$$

$$\mathbf{V} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

Compute the forward pass and calculate the network outputs $\hat{y}(t=1)$ and $\hat{y}(t=2)$:

- $\hat{y}(t=1) =$ ✓
- $\hat{y}(t=2) =$ ✓

The answers are integer numbers.

Frage 19

Nicht beantwortet

Nicht bewertet

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